# Restoration Optimization: Results/problems/debugging

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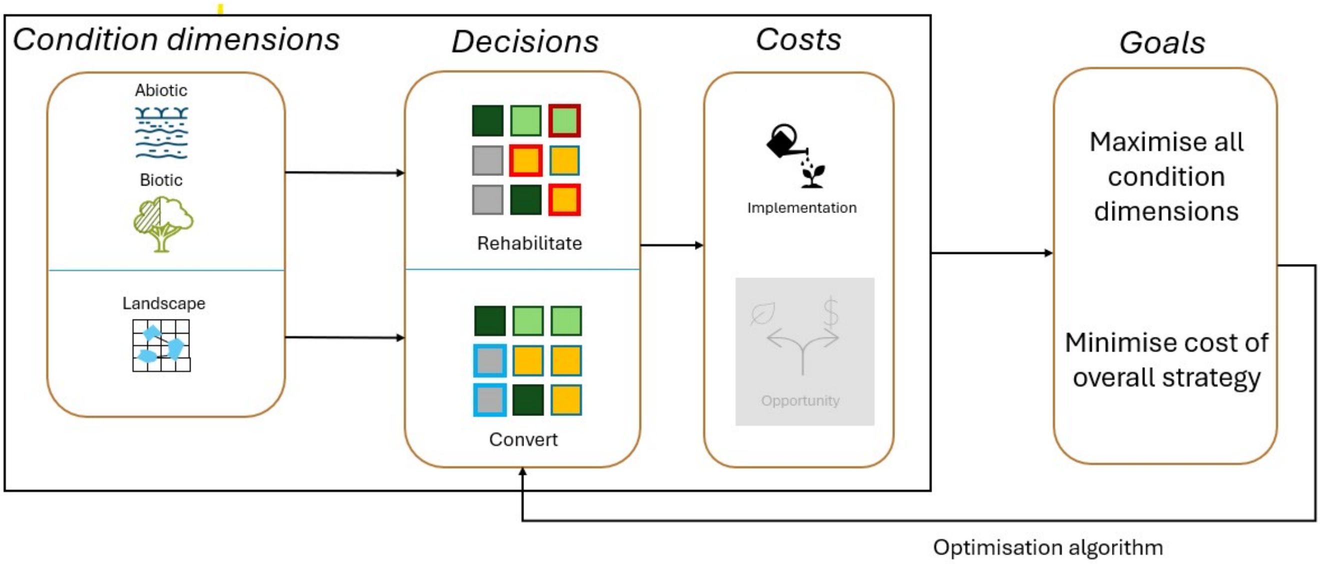
Code available at https://github.com/isabelntho/rest_prio

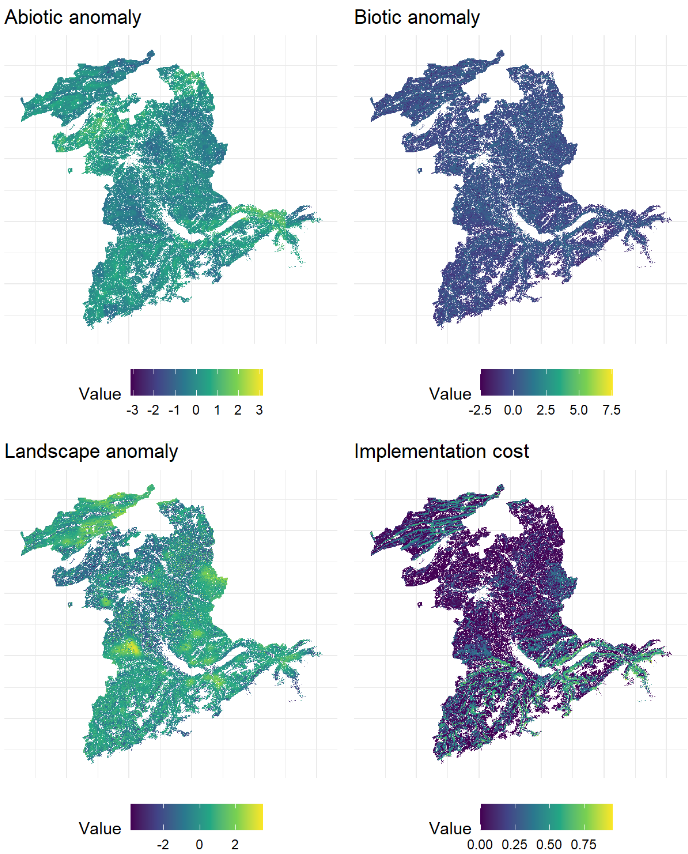
Can click on figures to make them larger

## Conceptual overview of workflow




### Initial Conditions

#### Spatial distribution



## Distribution of values


# Objectives & decisions

Decision to restore:

$$x_i^R \in \{0, 1\}$$

Decision to convert:

$$x_j^C \in \{0, 1\}$$

Within a set budget of actions

$$\sum_i x_i^R + \sum_j x_j^C = N_{\text{budget}}$$

## Overview of objective functions

### Abiotic & biotic condition

Positive or neutral anomalies receive no improvement. For restored pixels starting with a negative anomaly:

$$a_i^1 = a_i^0 + \alpha_A \, w(a_i^0)$$

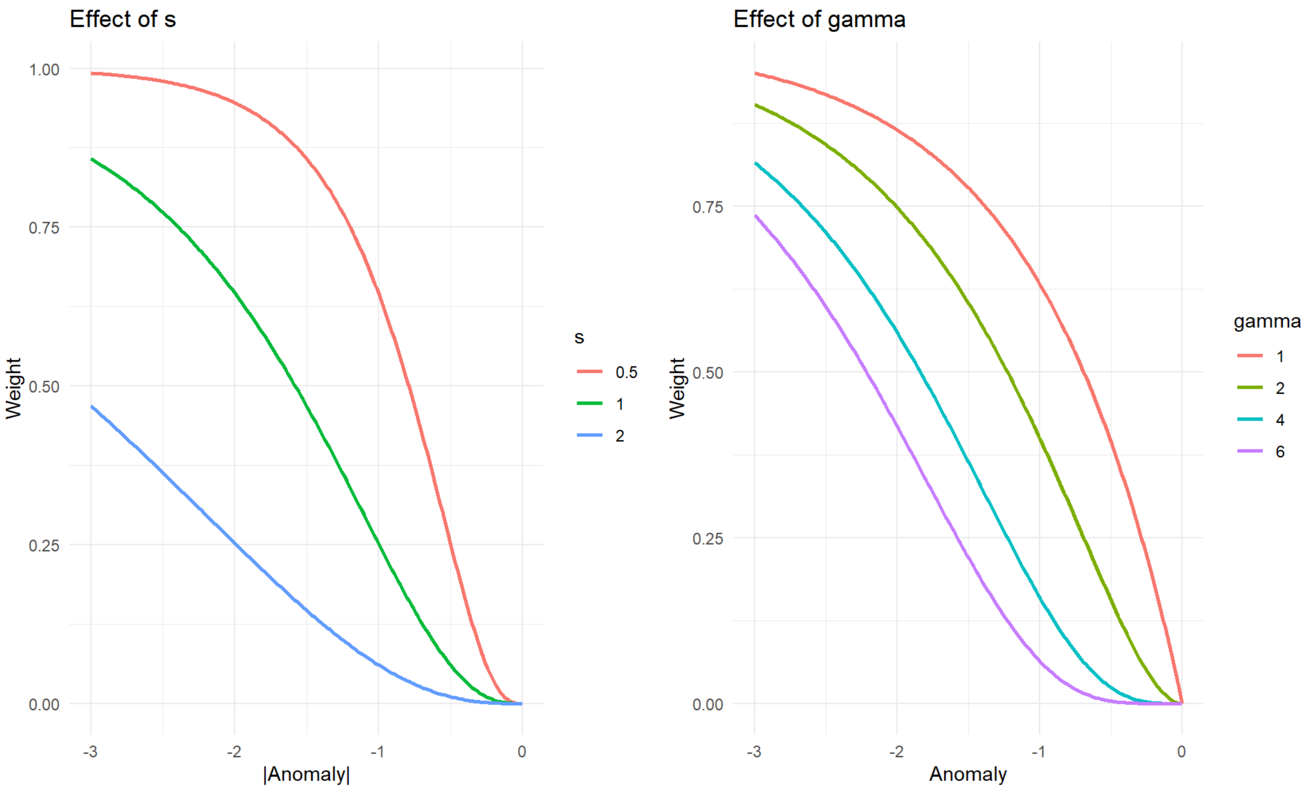
- $a_i^1$ is the anomaly after restoration, $a_i^0$ is the baseline anomaly

- $\alpha_A$ = abiotic effect, i.e. a constant that represents an arbitrary improvement achieved through restoration
- $w(a)$ is a weighting function based on the magnitude of the anomaly.

The improvement is scaled by a weight that depends on the baseline anomaly. This assigns larger improvement weights to pixels with larger negative anomalies. If the anomaly is close to zero, the weight is small. As the anomaly becomes more negative, the weight increases and approaches 1.

$$w(a) = \left(1 - e^{-\frac{|a|}{s}}\right)^{\gamma}$$

- s and $gamma$ are shape parameters that control how quickly the weight increases with the magnitude of the anomaly. Probably both aren't needed.




For pixels neighbouring restored pixels:

$$a_k^1 = a_k^0 + \alpha_A \, \lambda \, w(a_k^0)$$

- $\lambda$ is the effect decay with distance from the restored pixel.

Optimisation minimises the negative total improvement over the study area (i.e., Maximises total abiotic/biotic improvement from restoration):

$$f_{\text{a/biotic}}(x) = -\sum_{i \in \Omega_R} \left(a_i^1 - a_i^0\right)$$

Other choices for the objective function could be to maximise the improvement in the worst-off pixels (i.e., minimising the worst anomaly after restoration), or maximising the number of pixels for which condition passes a certain threshold.

**Landscape condition**

$x_j^C$ = 1 changes LULC of selected pixels to a desired class. Landscape anomaly is recalculated based on new LULC raster.

$$L^1 = 1 - D(\text{LULC}^1)$$

where

$$\text{LULC}^1 = \begin{cases} \text{focal class}, & \text{if } x_j^C = 1 \\ \text{LULC}^0, & \text{otherwise} \end{cases}$$

Optimisation minimises the total landscape anomaly across the whole study area:

Optimisation minimizes the total landscape anomaly across the whole study area.

$$f_{\text{landscape}}(x) = \sum_{p \in \Omega} L_p^1$$

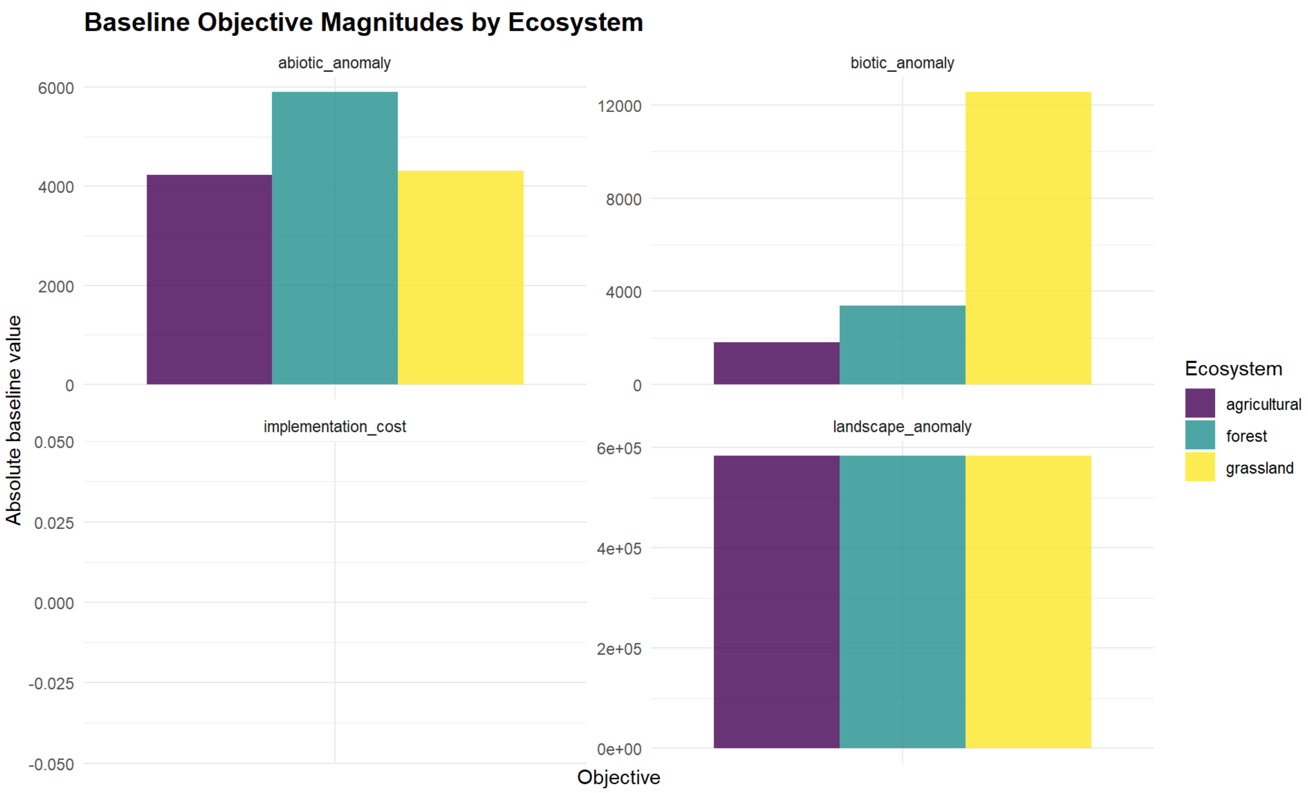
Optimisation minimizes the total landscape anomaly as the simple sum of the landscape anomaly raster values after applying the conversion driven landscape update.

**Implementation cost**

$$f_{\text{cost}}(x) = \sum_{k \in \Omega_R} y_k^R(x)\, c_k + \sum_{k \in \Omega_C} y_k^C(x)\, c_k$$

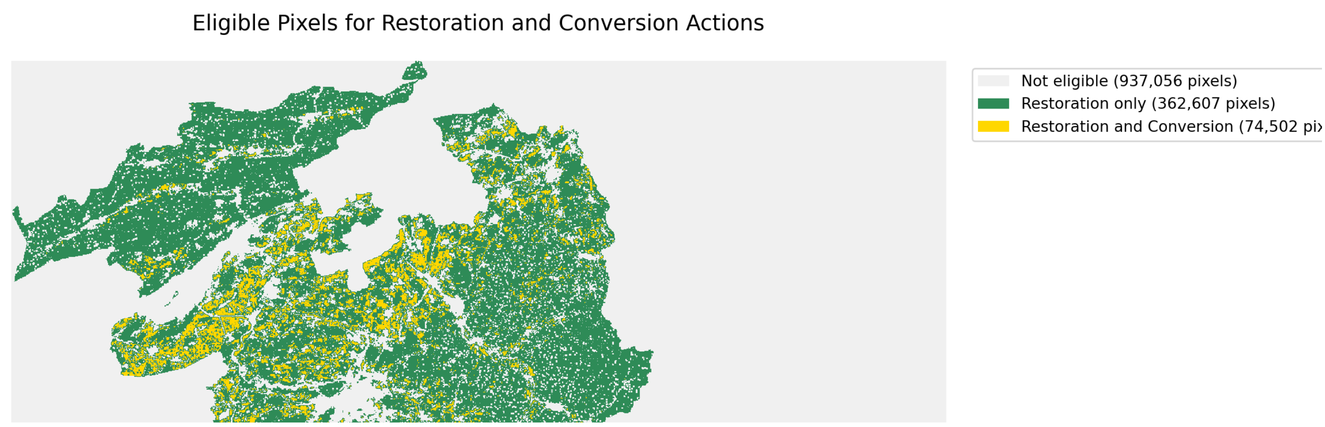
The code computes total implementation cost as the sum of baseline cost over all pixels selected for restoration plus all pixels selected for conversion, then minimizes that total.

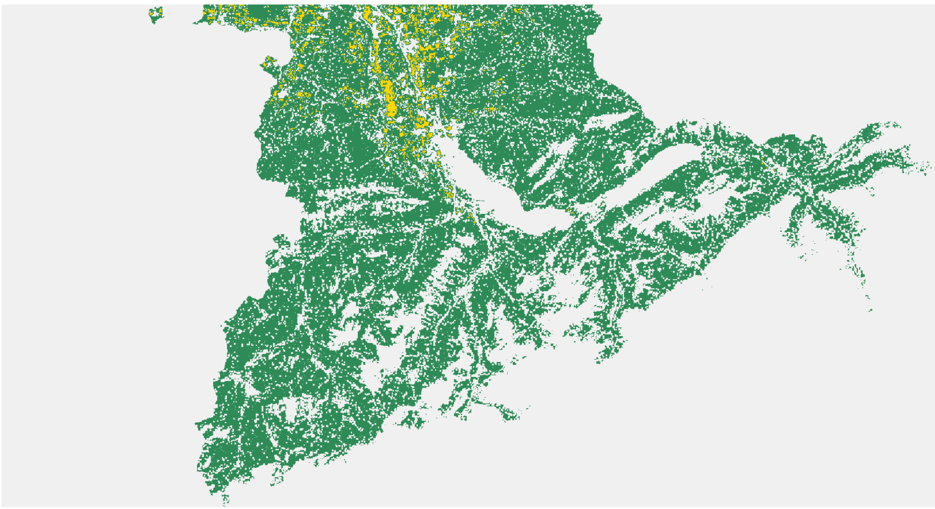
**Baseline values**




Calculation of the landscape anomaly same for each ecosystem type, because the calculation is the same…logically this might need some work. Implementation cost objective is 0 for all ecosystems, as cost only incurred after restoration decisions are applied.

## Eligible pixels for restoration and conversion





Pixels eligible for restoration: any pixels within the ecosystem type (Forest: Open & Closed Forest; Agricultural: Intensive Agriculture; Grassland: Grassland & Alpine Pastures)

## Constraints and repair

*Constraint formulation*

The optimisation is formulated with a fixed restoration budget.

The total number of selected pixels, including restoration and conversion actions, is constrained to equal a predefined maximum derived from the scenario specific restoration fraction.

*Repair strategy*

A unified adaptive repair operator is applied after crossover and mutation.

- Solutions that deviate strongly from the target number of actions (>20% of target number) are adjusted to restore feasibility.
- Small deviations are tolerated to preserve diversity.
- When repair is required, pixels are added or removed using a neutral, random mechanism.

# Output from optimisation runs

## Overview by Ecosystem

Show 10 entries Search:

| Metric | forest | agricultural | grassland |
|---|---|---|---|
| Objectives | 4 | 4 | 4 |
| Total_Pixels | 49567 | 28921 | 42481 |
| Max_Restored_Pixels | 4956 | 2892 | 4248 |
| Max_Restoration_fraction | 0.1 | 0.1 | 0.1 |
| Population_Size | 20 | 20 | 20 |
| Actual_Generations | 30 | 30 | 30 |
| Pareto_Solutions | 20 | 20 | 18 |

| | | | |
|---|---|---|---|
| Final_Hypervolume | 14812.07356807286 | 48.84298313658137 | 166.1217235265692 |

Summary by ecosystem

Showing 1 to 8 of 8 entries

Previous 1 Next

### Objective Statistics Comparison Across Ecosystems

Show 15 entries Search:

| Ecosystem | Objective | Min | Max | Mean | Std | Range | CV |
|---|---|---|---|---|---|---|---|
| All | All | All | All | All | All | All | All |
| forest | abiotic_anomaly | -2.8207 | -0.0607 | -1.6428 | 0.7848 | 2.7600 | 0.4777 |
| forest | biotic_anomaly | -8.5937 | -0.1859 | -4.6057 | 2.4186 | 8.4079 | 0.5251 |
| forest | landscape_anomaly | 1.2917 | 1.3001 | 1.2957 | 0.0025 | 0.0083 | 0.0019 |
| forest | implementation_cost | 48.0717 | 1,008.5867 | 508.7992 | 286.7973 | 960.5149 | 0.5637 |
| agricultural | abiotic_anomaly | -2.6546 | -0.0484 | -1.6659 | 0.7722 | 2.6062 | 0.4636 |
| agricultural | biotic_anomaly | -0.8992 | -0.0280 | -0.5696 | 0.2547 | 0.8712 | 0.4471 |
| agricultural | landscape_anomaly | 1.2953 | 1.2999 | 1.2978 | 0.0014 | 0.0047 | 0.0011 |
| agricultural | implementation_cost | 18.2450 | 24.7198 | 21.1812 | 1.7457 | 6.4749 | 0.0824 |
| grassland | abiotic_anomaly | -1.8818 | -0.5242 | -1.2246 | 0.4272 | 1.3576 | 0.3489 |
| grassland | biotic_anomaly | -0.2748 | -0.0846 | -0.1875 | 0.0612 | 0.1902 | 0.3263 |
| grassland | landscape_anomaly | 1.2943 | 1.3000 | 1.2971 | 0.0018 | 0.0056 | 0.0014 |
| grassland | implementation_cost | 48.3649 | 136.1754 | 92.1121 | 26.6714 | 87.8104 | 0.2896 |

Objective Statistics Comparison Across Ecosystems

Showing 1 to 12 of 12 entries

Previous 1 Next

### Decisions

(choice to restore vs convert)

Show 10 entries Search:

| Ecosystem | Restore_Min | Restore_Max | Restore_Mean | Restore_Std | Convert_Min | Con |
|---|---|---|---|---|---|---|
| forest | 61 | 4926 | 2,371.40 | 1,437.13 | 30 | |
| agricultural | 51 | 2766 | 1,547.10 | 823.25 | 126 | |
| grassland | 860 | 4149 | 2,462.44 | 1,059.42 | 99 | |

Decision Pattern Statistics by Ecosystem

Showing 1 to 3 of 3 entries

Previous 1 Next

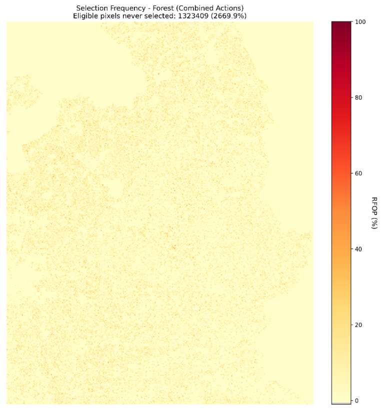
## Issues/debugging of optimisation results

# A: Weak spatial signal in selection

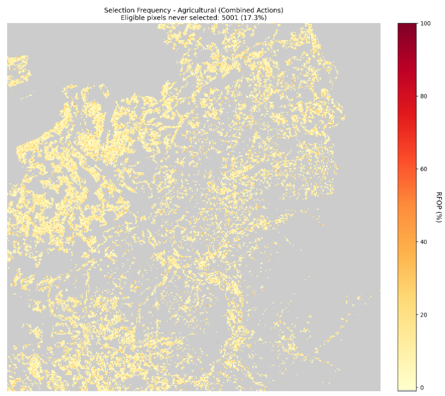
## Selection Frequency Maps

(small error in the pixel counts on these plots)

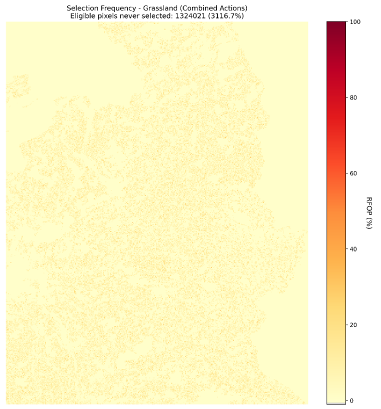
forest




agricultural



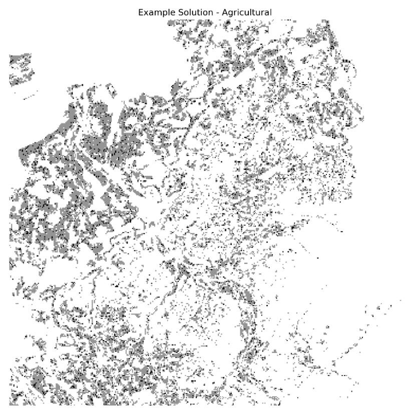

grassland




## Example Pareto-Optimal Solutions

forest


agricultural




grassland

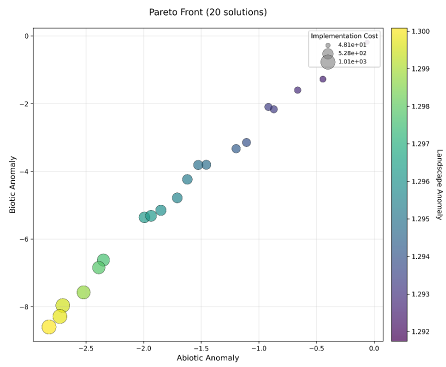

# B: Objective space problems

# B1: Redundancy of objectives

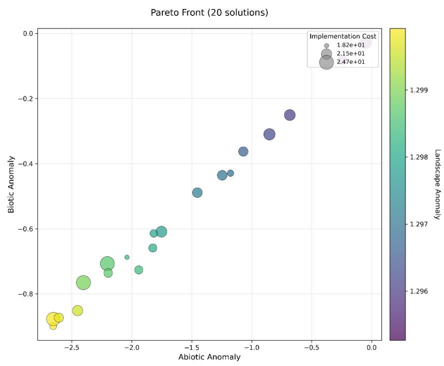
Cause identified - this is only because the landscape objective is not well implemented

## Pareto fronts

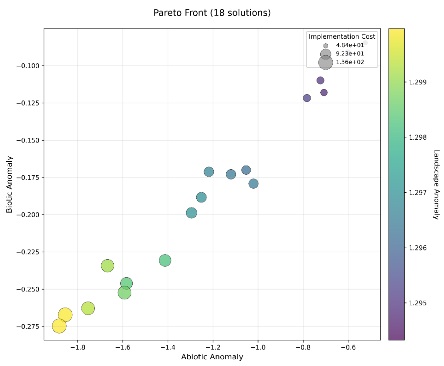
forest




agricultural

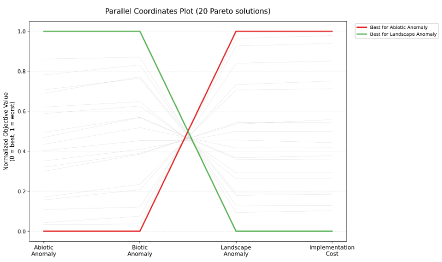



grassland

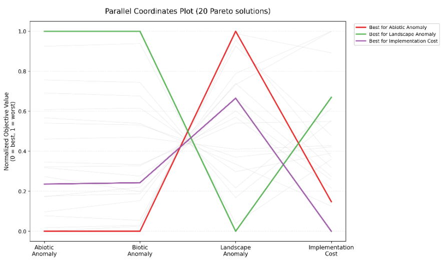


## Parallel coordinates

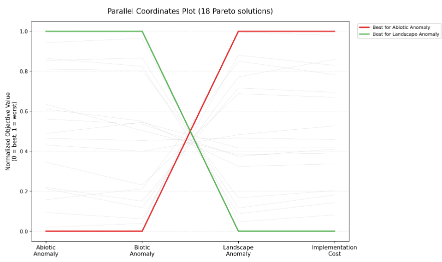
**forest**



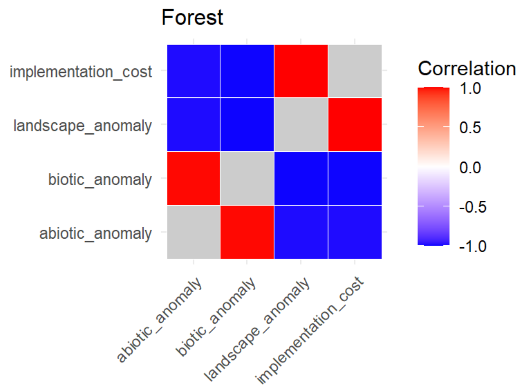

**agricultural**

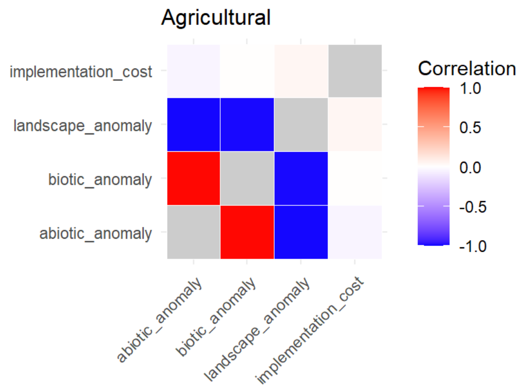


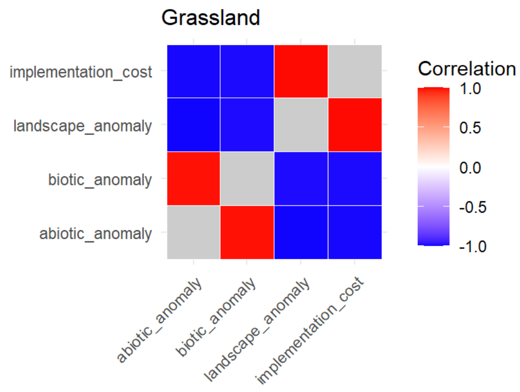

**grassland**




## Objective correlations



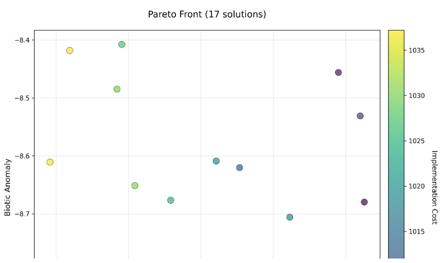




## Without landscape objective:

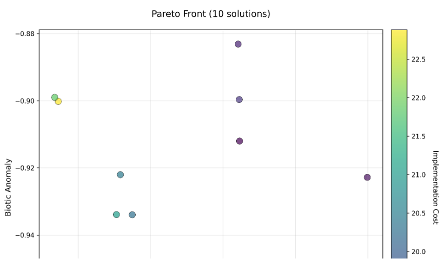
Running *without* landscape objective –> less solutions, higher diversity in solutions, weaker correlations between objectives

## Pareto fronts

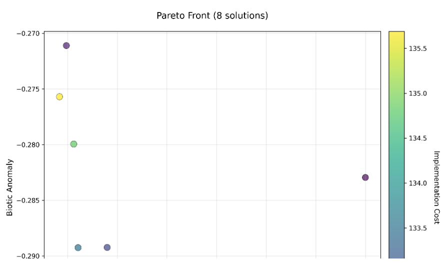
**forest**

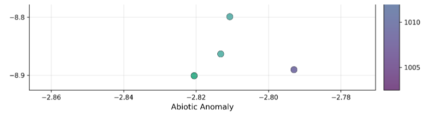


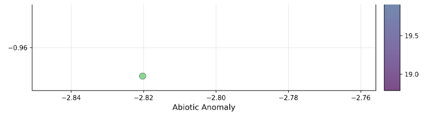

**agricultural**

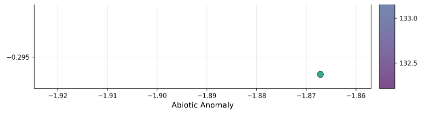



**grassland**



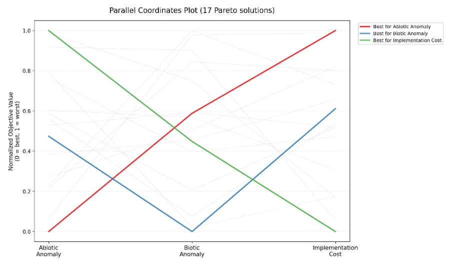


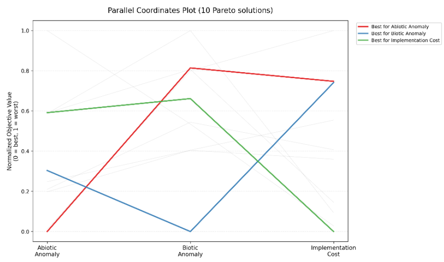




## Parallel coordinates

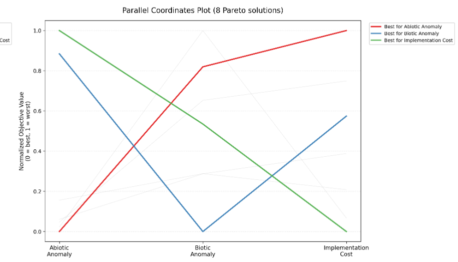
**forest**



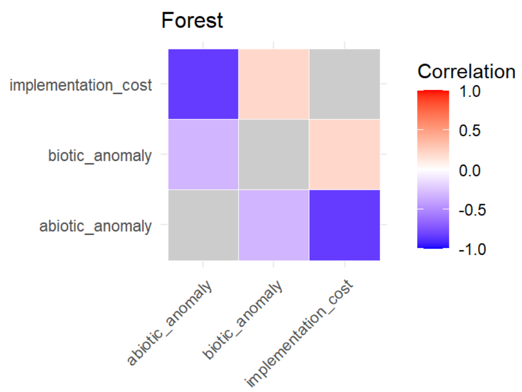

**agricultural**

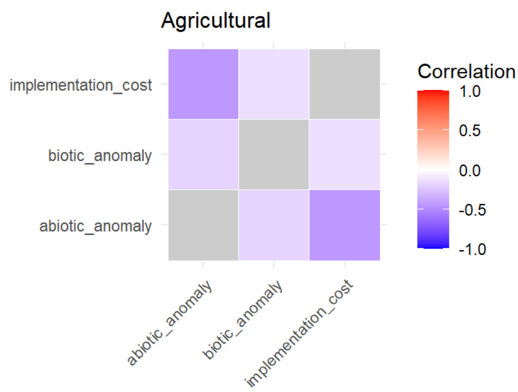


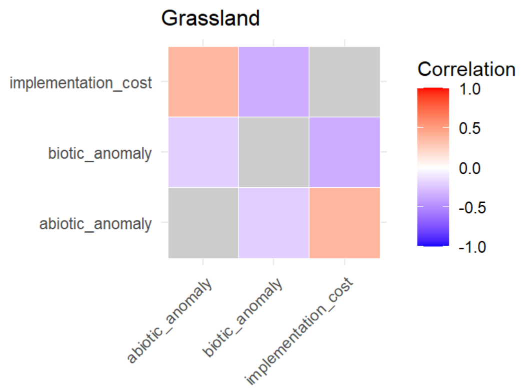

**grassland**




## Objective correlations



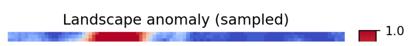



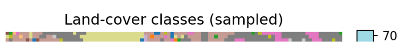

## B2: Weak sensitivity of landscape objective

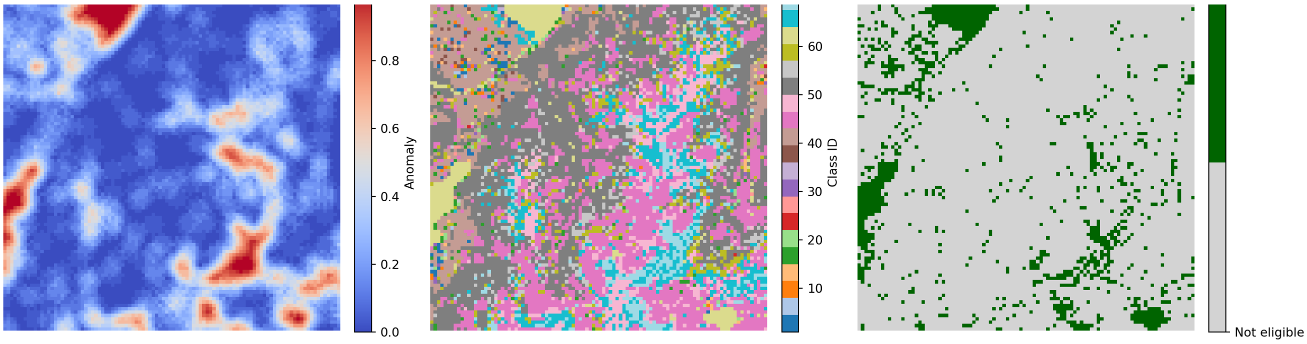
Basically the current set up of this objective drives very little change in landscape anomaly values across the solution space, so it has very little influence on selection of solutions. CV of landscape objective in all ecosystems is very low.

### Landscape Objective Sensitivity Test

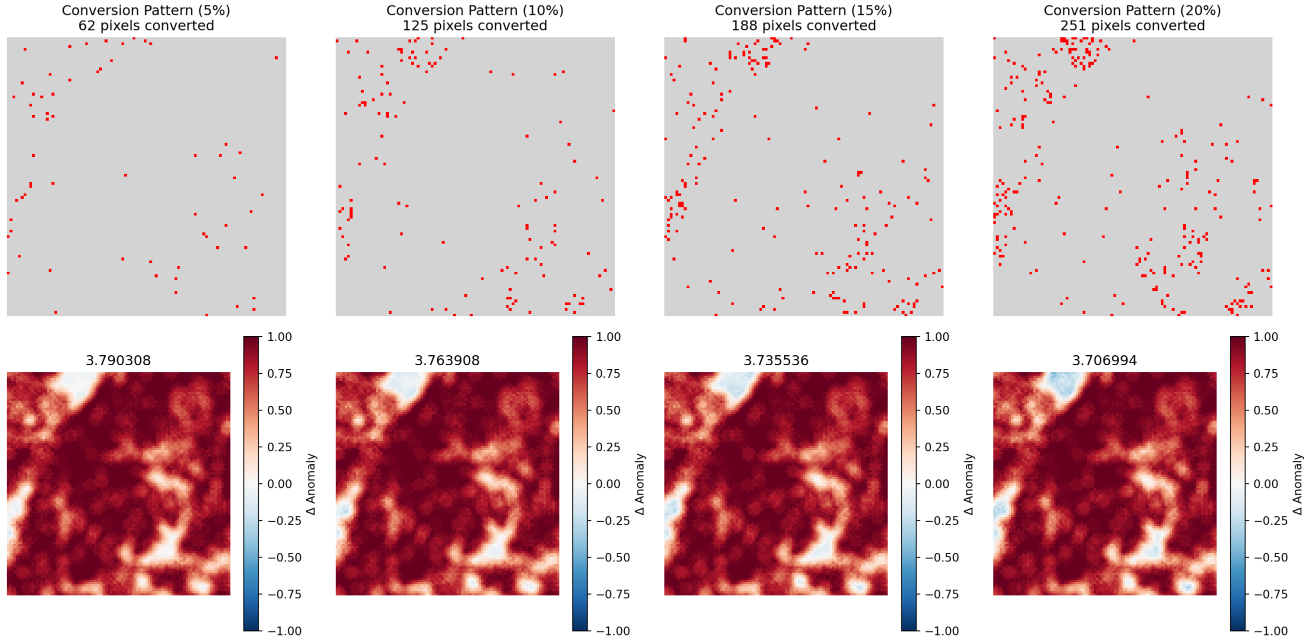
```
Sampled 100x100 area with 1259 eligible pixels for conversion
```

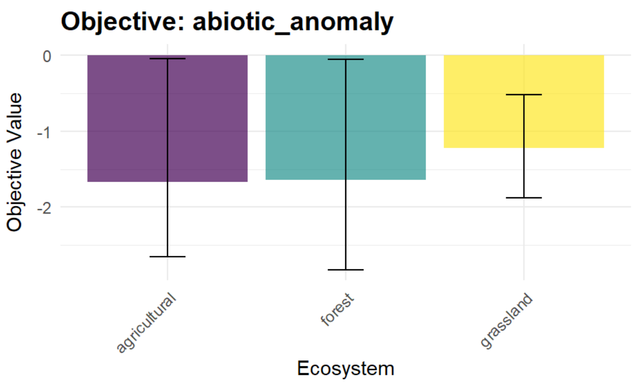


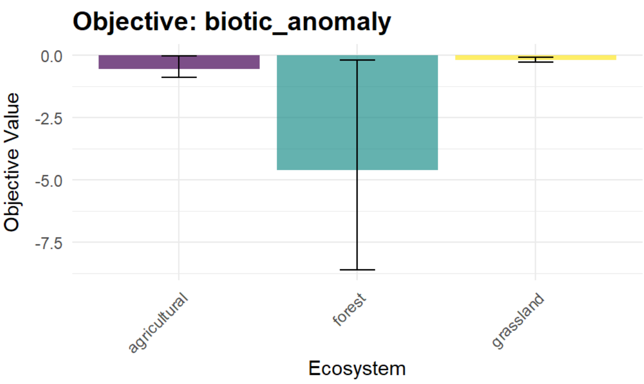



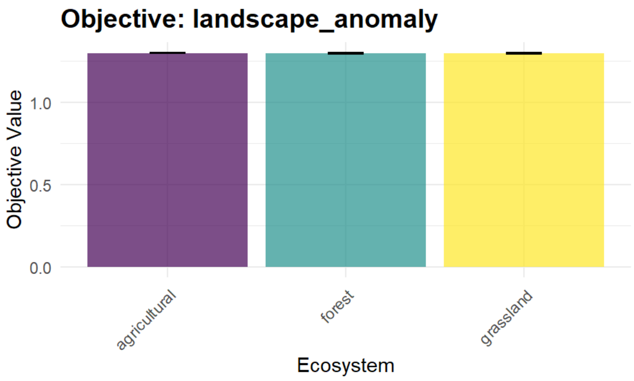



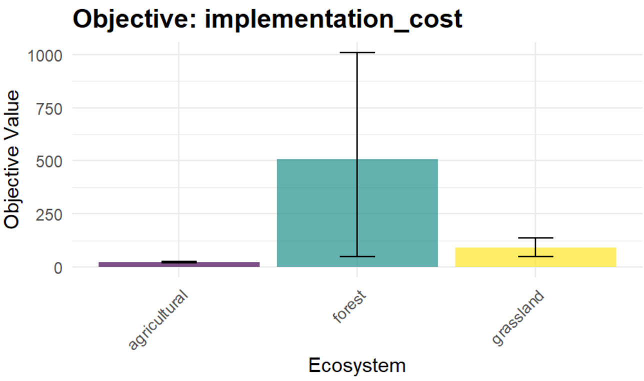

Different intensities of conversion produce only small changes to objective value/



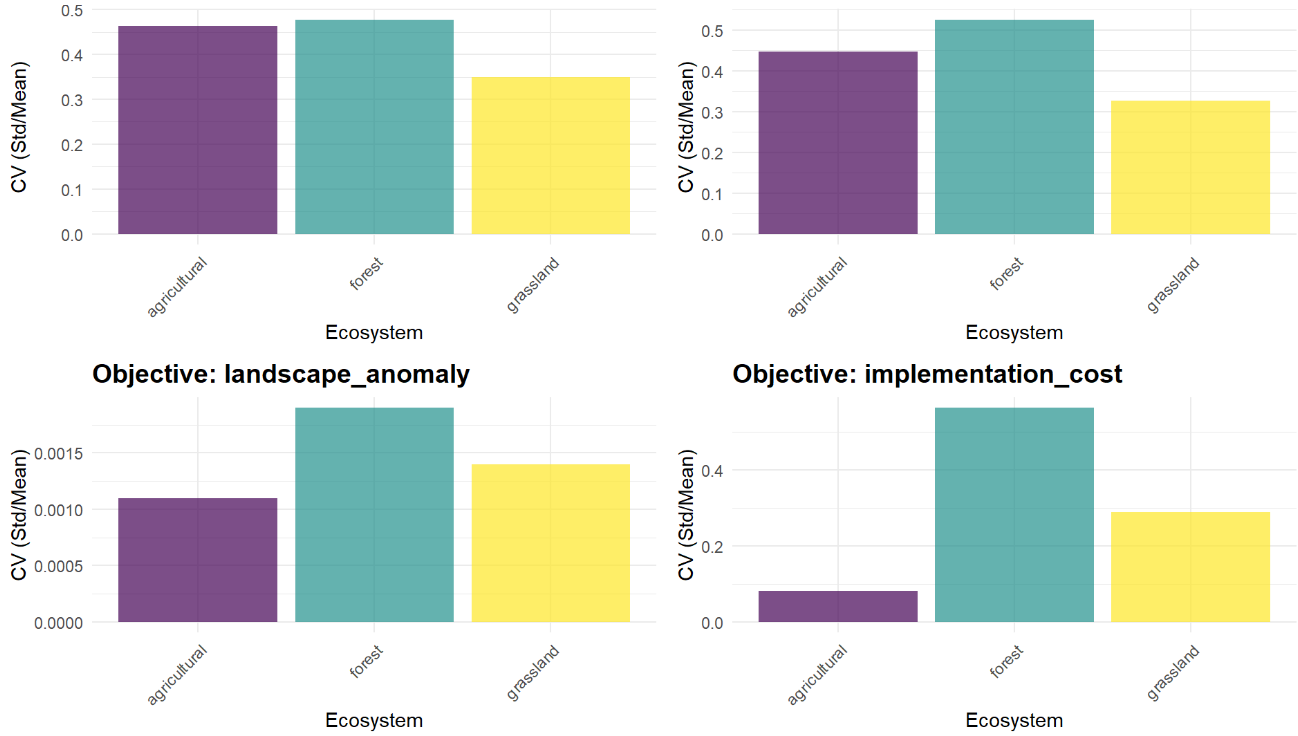



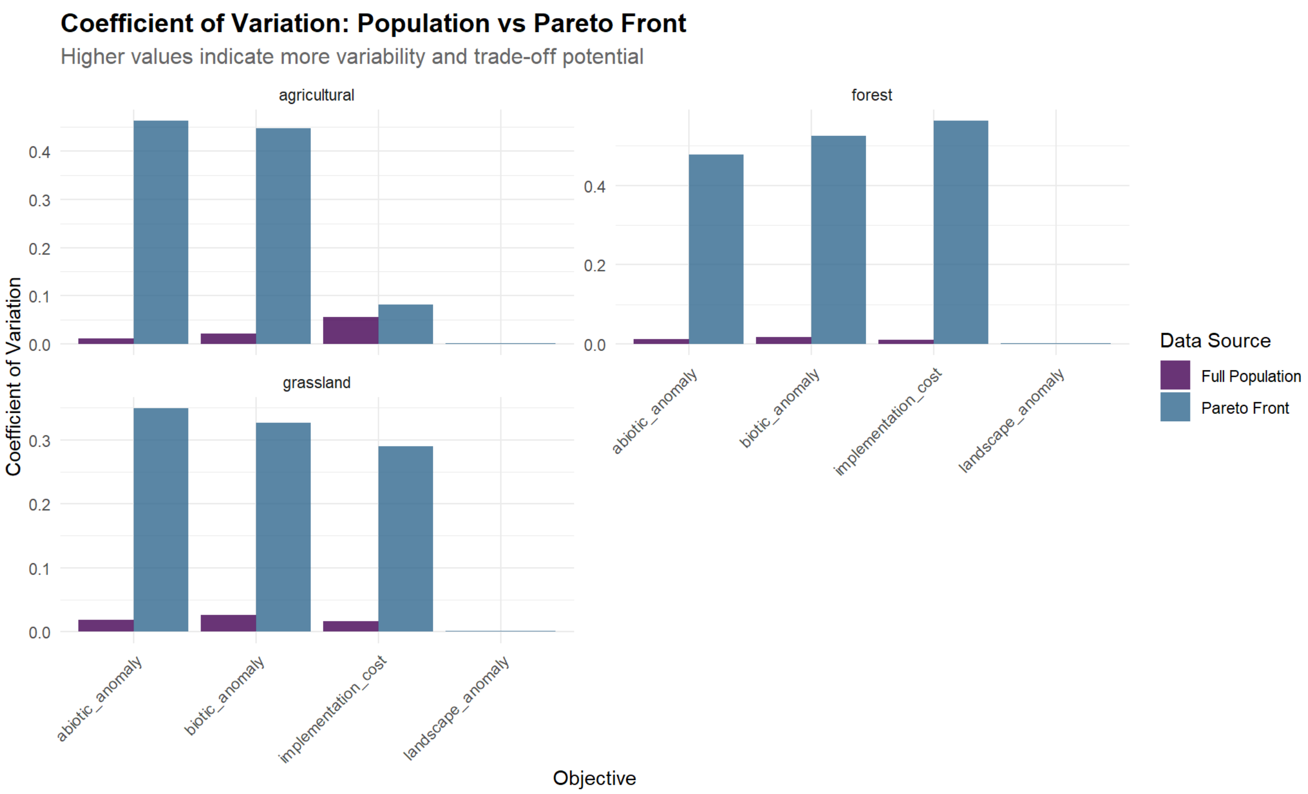

**Objective: abiotic_anomaly**

**Objective: biotic_anomaly**



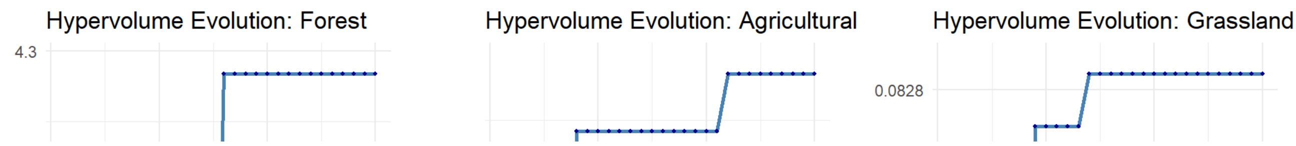

## Population vs Pareto Front Variance

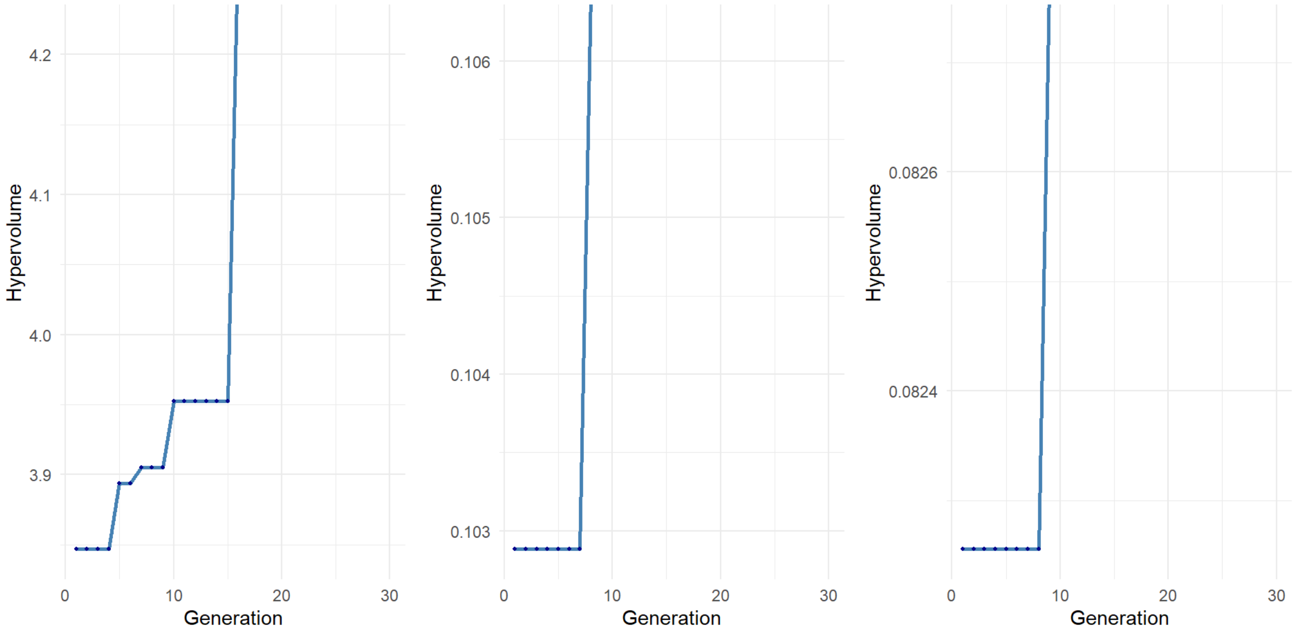
For run without landscape objective.



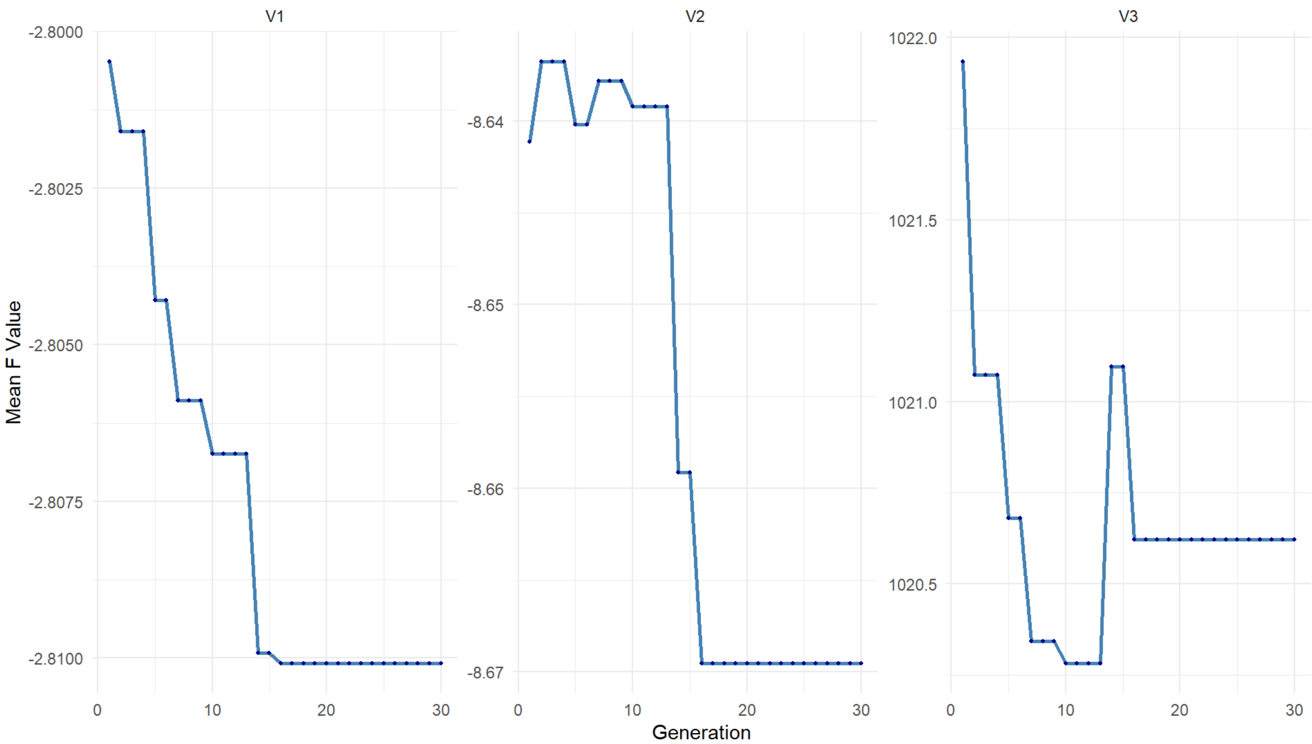

## Evolution Tracking

Hyperparameter and objective evolution, for run without landscape objective.

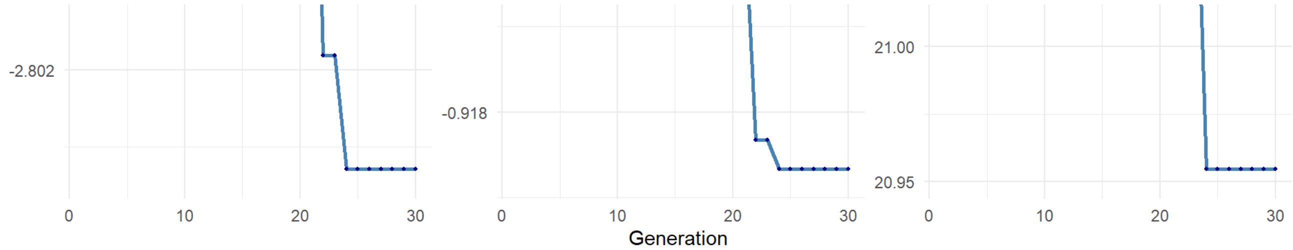





F Mean Evolution: Forest Ecosystem




F Mean Evolution: Agricultural Ecosystem




F Mean Evolution: Grassland Ecosystem